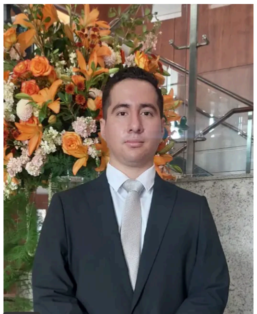
Daniel Alvarado Peláez



Degrees: Master in Cybersecurity | Mechatronic Engineer

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About me

I am an Ecuadorian **Mechatronics Engineer and Master in Cybersecurity**, passionate about technology and innovation. I have extensive experience in the design, development, and implementation of high-impact solutions, integrating **software development, cybersecurity, data analysis, and artificial intelligence** to solve complex challenges, with a focus on enhancing business productivity through ethical and efficient practices.

My profile combines a solid technical background with the ability to **lead technology teams and projects** throughout their entire lifecycle: from data ingestion to full-stack development, architecture, automation, and security audits. I am characterized by my analytical mindset, results-driven approach, and adaptability within **multidisciplinary and multilingual environments**, allowing me to successfully integrate into global and multicultural work dynamics.

Experience

General Specialist

Information Tecnology XOA S. A. / MAY 2023 - Present

- Full-stack development: design and implementation of secure web applications using modern technologies and coding best practices.
- Machine Learning model development and integration with company APIs and enterprise services, including production ML pipelines.
- Data analysis and processing: defining strategies for acquisition, cleaning, preprocessing, and handling large-scale data volumes with specialized tools.
- Information security: implementing protection measures, code audits, input validation, secure credential management, and service hardening.
- Security improvements: identifying vulnerabilities, assessing risks, designing mitigation solutions, and ensuring regulatory compliance.
- Technical support: field work for diagnostics, problem detection, and critical infrastructure issue resolution at client sites.
- Proactive monitoring: continuous supervision of services, servers, and production applications to ensure availability, performance, and security.
- Version updates and migrations: planning and implementing secure updates with minimal downtime and comprehensive validation.
- Internal tool development: creating company-specific services and utilities to optimize operational processes.

- Quality control: functional, security, and performance testing for embedded systems and web applications before release.

Research Assistant (Development Technician)

Center for Information Technologies (CTI) / MAR 2022 - APR 2022

- Participation in a research project focused on predicting the energy consumption of Data Center infrastructure using Machine Learning techniques.
- Design and development of Jupyter notebooks for data acquisition, cleaning, preprocessing, and exploratory data analysis from monitoring systems.
- Implementation, training, and comparative evaluation of different predictive models, using performance metrics to determine the most efficient solution.
- Optimization and improvement of a data collection application, enhancing the quality, consistency, and availability of information used in analytical models.
- Collaboration in research activities that contributed to the scientific publication presented at IEEE Future Networks World Forum (FNWF 2022).

Web Developer

Edu4Lab / MAY 2021 - JUL 2021

- Development of an educational web platform using Laravel under a scalable and maintainable system architecture.
- Implementation of user management, profile, and role modules, ensuring proper access control based on user responsibilities.
- Development of features for managing courses, books, online assessments, and academic staff.
- Design and integration of business logic and database for centralized management of educational content.
- Participation in functional testing and requirements validation to ensure proper platform operation.

Education

Postgraduate, Master in Cybersecurity

- Passed with honors; Cum laude

Universidad Casa Grande / Guayaquil, Ecuador / JUL 2024 - JAN 2026

Higher Education, Mechatronics Engineering

Escuela Superior Politécnica del Litoral (ESPOL) / Guayaquil, Ecuador / OCT 2017 - FEB 2023

Primary - Secondary Education

Unidad Educativa FAE Nº2 / Guayaquil, Ecuador / MAY 2005 - FEB 2017

Publications

Comparison of Traditional ML Algorithms for Energy Consumption Prediction Models

2022 IEEE Future Networks World Forum (FNWF), Montreal, QC, Canada, 2022, pp. 232-237, R. Estrada, V. Asanza, D. Torres, I. Valeriano and D. Alvarado

doi: 10.1109/FNWF55208.2022.00048.

Skills

Programming languages

JAVA / Python / C/C++ / HTML / JavaScript / PHP

Related

Node.js / Laravel / Jupyter Notebook / Dataiku / Astro / Vue.js

Databases

MySQL / Postgresql / MongoDB / Redis / Meilisearch

Related

Supabase / MongoDBCompass / MongoDB Atlas / Laragon / XAMPP / HeidiSQL / PGAdmin4 / DBeaver

Other development tools

Git & GitHub / Docker / Cloudflare / Copilot / MATLAB & Simulink

Cybersecurity

ISO 27001 / OWASP Top 10 / Personal Data Protection Law (Ecuador) / OSINT Tools / Shodan

Related

Vulnerability analysis / Risk management / Security audits

hardware tools

Arduino / Raspberry Pi / ESP32 - 8266 / PLC Siemens / PLC Logo

Other tools

Paquete Office / Notion / Odoo / Claude / Copilot / Canva

Technical skills

Development and programming / Analytics and data science / 3d design / Development of embedded systems

Related

Autodesk Inventor / Autodesk AutoCAD / Automation Studio / CADe Simu / CCW (Connected Components Workbench) / FluidSIM / LabVIEW / TIA Portal / Proteus / FlexSim

Personal skills

Teamwork / Commitment to tasks / Learning capacity / Orientation to results

Languages

Spanish - Native / English - C1

Projects

Thermal Monitoring Drone

CAD designs and Arduino circuits for a drone. CAD designs, circuits, and Raspberry Pi program for an embedded system that detects cable presence, captures thermal images, and detects temperature spikes.

Real-time OPC Simulation

Real-time circuit simulation, connected via OPC to a LABVIEW interface, for monitoring and control.

Proteus/Ubidots Pump Control

Design of a pump system in Proteus, controlled by an Atmega328p, and visualization in Ubidots of the system's status communicated by a program developed in Python.

Knee Prosthesis Adaptation

3D design of a knee and editing of a total knee prosthesis, using Slice 3D and Blender software to adapt the existing knee prosthesis to the test knee.

LABVIEW Pump Control

Design of an HMI for the control of a pump system, which has three pumps. Two pumps are primary and the third is backup in case one fails or is disabled for maintenance. Each pump has its start and stop pushbuttons, as well as its enable or disable selector. The system will have manual and automatic operation commanded by the signal sent by a selector.

Dynamic Mechanism Simulation

A mechanism is designed where the shaft of a mill receives the necessary power and speed through pulley-chains. In turn, the shaft that transmits power to the pulley with chain receives the input power through a set of V-belt pulleys, as well as being supported by 2 bearings.

Rice Stacker Design

Design and feasibility analysis of a rice stacker using renewable energy for implementation in a developing agricultural community.

Automatic Pet Feeder

Automatic pet feeder project controlled by a mobile application that allows scheduling feeding times.

Flexible Manufacturing Process

Redesigning the bottle manufacturing system of a company to reduce production times, increase system flexibility, and improve company profitability.

Sugar Production Quality

Objective:
Improve sugar production quality system by modifying manufacturing stages to avoid frequent product rejection upon delivery.
Issue:
A sugar-producing plant consists of several stages containing certain processes that make production efficient:
*Crystallization
*Juice delivery and extraction
*Evaporation
*Juice purification
The plant lacks sugar quality controls, so customers want quality tests based on random sampling so that if the product passes the tests, the order is accepted; otherwise, the product will be returned in full regardless of the order size.

Certificates

Despliegue de MySQL con Docker

MAR 2023

Python para data scientist avanzado

FEB 2023

Docker esencial

FEB 2023

Learning Docker

FEB 2023

OPENedX Escritura Académica

ENE 2023

Examen CAMBRIDGE ENGLISH PLACEMENT TEST, C1

DIC 2022

Cuarta revolución Industrial: Data science

FEB 2022

Aprende Excel (Office365/Microsoft365)

FEB 2022

Cómo eliminar las distracciones

FEB 2022

Cómo conciliar las funciones múltiples del líder

FEB 2022

Introducción a la programación en Python

ABR 2021

Fundamentos de programación en PLC

SEPT 2021

Python para data science y big data esencial

SEPT 2021

MATLAB Onramp

OCT 2020

Certificate of Competency in English, ECCE B2

NOV 2017

Certificado de graduación, CEN

MAY 2017

[Link to certificates]

References

Ph.D. Víctor Manuel Asanza - Professor of the FIEC faculty

Company/Institution: ESPOL University

Phone: +593 98 265 9680

Ing. Tamara García - Supervisor

Company/Institution: Delta Delfini

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Mgs. Alexandra Pérez Rivera - Tourism Course Teacher

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Darwin León - Manager

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